

□ Solve $\int \frac{dx}{\sqrt{x^2-1}}$ (from 6.14 Example 2.c.)

Let $\theta = \operatorname{arcsec} x \Rightarrow x = \sec \theta$, $\{\theta \in \mathbb{R} \mid \theta \neq \frac{\pi}{2} + n\pi, n \in \mathbb{Z}\}$

~~we limit~~ First, consider the domain of x :

$$x^2 - 1 > 0 \Rightarrow x^2 > 1 \Rightarrow x > 1 \text{ or } x < -1$$

Note that $x \neq \pm 1$, thus $\sec \theta \neq \cos \theta \neq \pm 1 \Rightarrow \theta \neq 0, \pi$

For simplicity, we limit θ to $[0, \pi)$.

Therefore, after considering all restrictions, $\theta \in (0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi)$.

$$\frac{dx}{d\theta} = \frac{d}{d\theta} \sec \theta = \sec \theta \tan \theta$$

$$dx = \sec \theta \tan \theta d\theta$$

Substitute $dx = \sec \theta \tan \theta d\theta$ and $x = \sec \theta$ into the original integral:

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2-1}} &= \int \frac{\sec \theta \tan \theta d\theta}{\sqrt{\sec^2 \theta - 1}} = \int \frac{\sec \theta \tan \theta d\theta}{\sqrt{\tan^2 \theta}} \\ &= \int \frac{\sec \theta \tan \theta d\theta}{|\tan \theta|} \end{aligned}$$

This is a piecewise function, or integral:

$$\begin{cases} \int \frac{\sec \theta \tan \theta d\theta}{\tan \theta}, & \theta \in (0, \frac{\pi}{2}) \\ \int \frac{\sec \theta \tan \theta d\theta}{-\tan \theta}, & \theta \in (\frac{\pi}{2}, \pi) \end{cases}$$

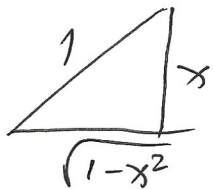
$$= \begin{cases} \int \sec \theta d\theta, & \theta \in (0, \frac{\pi}{2}) \\ \int -\sec \theta d\theta, & \theta \in (\frac{\pi}{2}, \pi) \end{cases}$$

$$\begin{cases} \int \sec \theta d\theta, \theta \in (0, \frac{\pi}{2}) \\ \int \sec \theta d\theta, \theta \in (\frac{\pi}{2}, \pi) \end{cases} = \begin{cases} \ln |\tan \theta + \sec \theta| + C_1, \theta \in (0, \frac{\pi}{2}) \\ -\ln |\tan \theta + \sec \theta| + C_2, \theta \in (\frac{\pi}{2}, \pi) \end{cases}$$

Substitute $\theta = \operatorname{arcsec} x$ back,

$$\begin{cases} \ln |\tan(\operatorname{arcsec} x) + \sec(\operatorname{arcsec} x)| + C_1, x \in (1, \infty) \\ -\ln |\tan(\operatorname{arcsec} x) + \sec(\operatorname{arcsec} x)| + C_2, x \in (-\infty, -1) \end{cases}$$

Consider $\tan(\operatorname{arcsec} x)$; for a triangle whose hypotenuse is 1 and x as a leg: $\operatorname{arcsec} x = \frac{1}{\cos} \theta \Rightarrow \sec \theta = x$



~~$$\frac{1}{\cos \theta} = x \Rightarrow \cos \theta = \frac{1}{x}$$~~

$$\text{From } \tan^2 \theta + 1 = \sec^2 \theta$$

$$\tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{x^2 - 1}$$

Therefore, $\begin{cases} \ln |\tan(\operatorname{arcsec} x) + \sec(\operatorname{arcsec} x)| + C_1, x \in (1, \infty) \\ -\ln |\tan(\operatorname{arcsec} x) + \sec(\operatorname{arcsec} x)| + C_2, x \in (-\infty, -1) \end{cases}$

$$= \begin{cases} \ln |\sqrt{x^2 - 1} + x| + C_1, x \in (1, \infty) \\ -\ln |\sqrt{x^2 - 1} + x| + C_2, x \in (-\infty, -1) \end{cases}$$

$$\text{Ans: } \int \frac{dx}{\sqrt{x^2 - 1}} = \begin{cases} \ln |\sqrt{x^2 - 1} + x| + C_1, x \in (1, \infty) \\ -\ln |\sqrt{x^2 - 1} + x| + C_2, x \in (-\infty, -1) \end{cases}$$